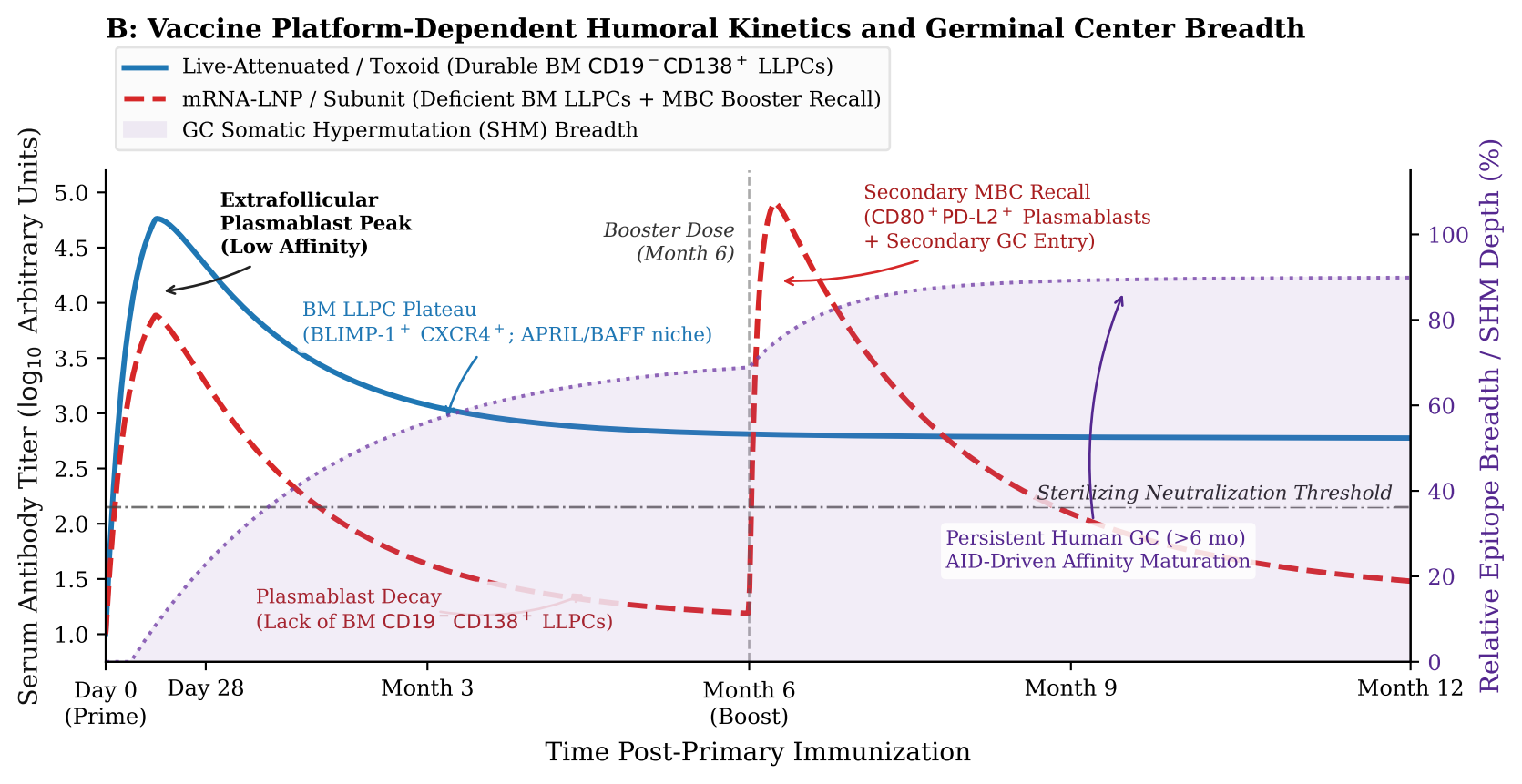

B: Vaccine Platform-Dependent Humoral Kinetics and Germinal Center Breadth
Live-Attenuated / Toxoid (Durable BM CD19⁻CD138⁺ LLPCs)
mRNA-LNP / Subunit (Deficient BM LLPCs + MBC Booster Recall)
GC Somatic Hypermutation (SHM) Breadth
Extrafollicular Plasmablast Peak (Low Affinity)
Booster Dose (Month 6)
Secondary MBC Recall (CD80⁺PD-L2⁺ Plasmablasts + Secondary GC Entry)
BM LLPC Plateau (BLIMP-1⁺ CXCR4⁺; APRIL/BAFF niche)
Sterilizing Neutralization Threshold
Persistent Human GC (>6 mo) AID-Driven Affinity Maturation
Plasmablast Decay (Lack of BM CD19⁻CD138⁺ LLPCs)
Serum Antibody Titer (log10 Arbitrary Units)
Relative Epitope Breadth / SHM Depth (%)
Day 0 (Prime)
Day 28
Month 3
Month 6 (Boost)
Month 9
Month 12
Time Post-Primary Immunization